

Assignment -1

(1) 4-bit binary adder

$$S_i = A_i \oplus B_i \oplus C_i$$

$$C_{i+1} = A_i B_i + C_i (A_i \oplus B_i)$$

(2) 3-bit Async JK counter

Set $J = K = 1$ for all ff.

000 $\rightarrow$ 001 $\rightarrow$ 010 $\rightarrow$ 011 $\rightarrow$ 100 $\rightarrow$ 101

↓
000 $\leftarrow$ 111 $\leftarrow$ 110

3) Hazards in combinational
Circuits.

Static-1 hazard

Static-0 hazard

Dynamic hazard.

Elimination: add a redundant/
consensus term to the boolean
expression or use hazard free
circuit.

101 $\rightarrow$ 001 $\rightarrow$ 110 $\rightarrow$ 010 $\rightarrow$ 100 $\rightarrow$ 000

$\downarrow$

011 $\rightarrow$ 111 $\rightarrow$ 000

4) Setup & hold time at 100 MHz

$$\text{freq } f = 100\text{ MHz}$$

$$\text{Clock period } T = \frac{1}{100\text{ MHz}} = 10\text{ ns}$$

for correct operation

$$T_{\text{clk}} \geq T_{\text{eq}} + T_{\text{comt}} + T_{\text{setup}}$$

&

$$T_{\text{eq}}(\text{min}) + T_{\text{comt}}(\text{min}) \geq T_{\text{hold}}.$$